

Week Four: Centrality Measures

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DATA 620: Web Analytics

Research question

How does the degree centrality of subway shared-service networks vary by borough, and might degree centrality be used to predict higher station ridership?

I selected this topic because of my interest in transportation, specifically how network connectedness relates to passenger flow. My proposal outlines the data source, methodology, and possible outcome. No data has been collected or analyzed yet.

Data source

I would use the **MTA Subway Stations** data set provided by New York State Open Data. It contains station details, borough, and weekday daytime lines. The web API satisfies the data access criterion of the assignment.

- [Dataset page](#)
- [Socrata JSON endpoint](#)

Station ID, station name, borough, and daytime routes would be retained. The categorical variable to use would be **borough**, whereby comparison will be made for Manhattan, Brooklyn, Queens, and the Bronx. Staten Island Railway would be left out of this analysis.

Network definition

The network will comprise nodes, each representing a station ID. An edge would exist between two nodes when their respective daytime-route lists intersect at least with one common route. The network will be unweighted and undirected with no self-loops and multiple edges. The borough would be a node attribute.

It is a **shared-service network**. The presence of an edge does not mean that stations are neighboring each other on the railroad tracks or that each scheduled train visits both stations. Branches and services make such differentiation necessary. Stations within one transfer complex are considered separate stations if they have different station IDs.

Degree is calculated as the number of other stations having at least one common route to the station. Degree centrality is divided by the total number of other stations. A station located on a long route can have a high centrality value despite being part of one route only.

High-level analysis plan

1. **Retrieve the records.** I will fetch the JSON using Python **requests** library, validate the request, paginate through it till all the records are fetched. I will document the date of fetch since service assignments might have changed by then.

2. **Data Preparation.** I will create a **pandas** DataFrame, keep necessary fields, check for duplicated station IDs and normalize the borough names. Missing boroughs or routes, if any, will be validated first before being discarded.
3. **Graph Visualization.** I would convert the route list of each station to a set. Using the package **NetworkX**, I would introduce the eligible stations along with their borough data, connecting station pairs that have overlapping route sets. Introducing the nodes first would take care of the isolated stations.
4. **Centralities calculation.** I would calculate the degree centrality of the complete graph and merge it back with the station dataset. The same denominator of the whole graph will be used by every borough. I would not calculate the subgraphs of the boroughs separately for this comparison.
5. **Comparison of groups.** I would describe the number of stations, average centrality, and median centrality per borough. Boxplot would visualize the variability of every group, and ranking table would reveal the top central stations. Betweenness centrality may be calculated for additional comparison.

Hypothetical outcome

My hypothesis is that the degree centrality of the stations in Manhattan will be greater on average compared to other borough stations since some of them might be served by a larger number of stations. In addition, I will hypothesize that higher-centrality stations will have higher ridership. This is a hypothesis, not an actual finding.

In order to validate the hypothesis about the correlation of the centrality and ridership, I would get another dataset from MTA with information on ridership and make sure that its period is aligned with the station service dataset. If ridership is calculated in station complexes, I would construct a network with complex nodes and calculate centrality again. I will not allocate the ridership of a complex evenly among all its stations.

Service frequency, length of the route, population density around, employment, and touristic sites could affect ridership. All these variables should be controlled in order to see if centrality adds any predictive power. Correlation of centrality and ridership does not prove causation.

Reference

Metropolitan Transportation Authority. *MTA Subway Stations*. New York State Open Data. <https://data.ny.gov/Transportation/MTA-Subway-Stations/39hk-dx4f>